Functional Annotation Report: FolD2 (Q88KM5, PP_2265) in *Pseudomonas putida* KT2440

Gene/Protein Identity — Verified

Field	Value
UniProt	Q88KM5
Gene	*folD2* (syn. *folD-2*); ordered locus **PP_2265**
Organism	*Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125), NCBI taxid 160488
Length / MW	284 aa / 30.6 kDa
Protein	Bifunctional protein FolD 2
EC numbers	1.5.1.5 (dehydrogenase) + 3.5.4.9 (cyclohydrolase)
Family	Tetrahydrofolate dehydrogenase/cyclohydrolase family (HAMAP MF_01576)
Domains	THF_DH/CycHdrlase catalytic (IPR020630); NAD(P)-binding (IPR020631, IPR036291); Aminoacid_DH-like N (IPR046346)

Identity confirmed. The gene symbol *folD2* matches the protein description (bifunctional methylenetetrahydrofolate dehydrogenase / methenyltetrahydrofolate cyclohydrolase), the organism is correct, and the domain architecture (an N-terminal folate-binding domain plus a Rossmann-like NAD(P)-binding domain) is exactly that of the FolD family. Literature on bacterial FolD (E. coli, *P. aeruginosa*, *A. baumannii*, *C. perfringens*) is directly applicable. Note: *P. putida* KT2440 carries **two** FolD paralogs — *folD1* (Q88LI7, 291 aa) and *folD2* (Q88KM5, PP_2265, 284 aa) — a duplication not present in *E. coli*.

Quantitative homology transfer (own sequence analysis). Global pairwise alignment (Needleman–Wunsch) shows FolD2 is **86.6% identical** to the crystallographically and kinetically characterized *P. aeruginosa* PAO1 FolD (Q9I2U6, identical 284-aa length; structure in P 22558288) and **67.3% identical** to the biochemically characterized *E. coli* FolD (P24186). By contrast, the *P. putida* paralog FolD1 shares only **53.5%** identity with FolD2 and is more divergent from characterized orthologs (~53% *E. coli*, ~55% *P. aeruginosa*). Thus **FolD2 is the close, canonical FolD ortholog**, and structural/kinetic conclusions from *P. aeruginosa* and *E. coli* FolD transfer to FolD2 with high confidence; FolD1 is the more divergent copy.

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1. Summary (Answer to the Research Question)

FolD2 is a **soluble, cytoplasmic, $NADP^+$-dependent bifunctional enzyme** of folate (one-carbon) metabolism. As a homodimer it catalyzes two consecutive reactions on the tetrahydrofolate (THF) scaffold: (i) $NADP^+$-dependent **oxidation of 5,10-methylene-THF to 5,10-methenyl-THF** (methylenetetrahydrofolate dehydrogenase, EC 1.5.1.5), and (ii) **hydrolysis of 5,10-methenyl-THF to 10-formyl-THF** (methenyltetrahydrofolate cyclohydrolase, EC 3.5.4.9). Its product, 10-formyl-THF, is the one-carbon donor for **de novo purine biosynthesis** and for **formylation of initiator Met-tRNA^fMet**, while the interconverted folate pool also supports thymidylate, methionine and histidine metabolism. The enzyme acts in the cytoplasm as a central housekeeping node distributing one-carbon units.

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2. Primary Function: The Reactions Catalyzed

FolD interconverts oxidized one-carbon-loaded folates through two tightly coupled steps in a single active-site cleft:

1. **Dehydrogenase (EC 1.5.1.5):** (6R)-5,10-methylene-THF + $NADP^+$ → (6R)-5,10-methenyl-THF + NADPH
2. **Cyclohydrolase (EC 3.5.4.9):** (6R)-5,10-methenyl-THF + H_2O → (6R)-10-formyl-THF + H^+

The net effect is oxidation of the methylene one-carbon unit up to the formyl oxidation level, generating both NADPH and 10-formyl-THF. "Most organisms possess bifunctional FolD ... to generate NADPH and 10-formyltetrahydrofolate (10-CHO-THF) required in various metabolic steps" (Aluri et al., 2016, P 26531681), and FolD "catalyzes interconversion of 5,10-methylene-tetrahydrofolate and 10-formyl-tetrahydrofolate in the one-carbon metabolic pathway" (Sah & Varshney, 2015, P 25988590).

Substrate specificity / cofactor. The folate substrate is polyglutamylated tetrahydrofolate bearing a one-carbon unit at N5/N10; the dehydrogenase step is specific for **$NADP^+$** (not NAD^+) in this HAMAP-annotated FolD. The reaction is reversible in vitro but physiologically drives methylene→formyl flux.

Mechanistic separability. Structure-guided mutagenesis of *E. coli* FolD shows the two activities are catalytically distinct yet co-located: "the R191E mutant showed high cyclohydrolase activity, it retained only a residual dehydrogenase activity ... the K54S mutant lacked the cyclohydrolase activity but possessed high dehydrogenase activity" (Sah & Varshney, 2015, P 25988590). Residues D121/G122 are shared and essential for both. This supports a shared cleft between the two domains with the labile 5,10-methenyl-THF intermediate channeled from the dehydrogenase to the cyclohydrolase site.

Active site is intact in FolD2 (own residue-level analysis). Alignment of FolD2 to *E. coli* FolD shows **all seven** functionally characterized residues are conserved at identical positions in FolD2: **Y50, K54, C58, Q98, D121, G122, R191**. Critically, the cyclohydrolase-specific **K54** and the dehydrogenase-specific **R191** are both present. This provides direct residue-level evidence that FolD2 is a **genuinely bifunctional** enzyme retaining both EC 1.5.1.5 and EC 3.5.4.9 activities — not a monofunctional or degenerate paralog (contrast with *C. perfringens*, where FolD is dehydrogenase-only and a separate FchA supplies cyclohydrolase activity; P 26531681).

3. Localization

FolD2 is a **cytoplasmic** enzyme. It has no signal peptide, no transmembrane segment, and no membrane-anchoring motif; UniProt annotates it as a soluble **homodimer**. Its substrates (folates, $NADP^+$) are cytosolic metabolites, and all characterized bacterial FolD orthologs are soluble cytoplasmic proteins purified in recombinant systems (e.g., *P. aeruginosa*, *A. baumannii*; PMIDs 22558288, 23050773). It therefore performs its function within the bacterial cytoplasm as part of the soluble one-carbon metabolic machinery.

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4. Pathway Context and Biological Processes

FolD occupies a branch point of the **tetrahydrofolate one-carbon cycle**. 5,10-methylene-THF is generated mainly from serine by serine hydroxymethyltransferase (GlyA) and from glycine cleavage. FolD oxidizes this pool to **10-formyl-THF**, which is consumed by:

- **De novo purine biosynthesis** — two transformylase steps (GAR and AICAR transformylases) that build the purine ring (UniProt keyword: Purine biosynthesis).
- **Initiator tRNA formylation** — 10-formyl-THF is the formyl donor for methionyl-tRNA formyltransferase (Fmt), producing fMet-tRNA^fMet to prime translation. FolD is genetically tied to this process: resistance to the peptide deformylase inhibitor actinonin arises from mutations in "the fmt gene … and the folD gene encoding the bifunctional enzyme methylenetetrahydrofolate-dehydrogenase and -cyclohydrolase" (Nilsson et al., 2006, P 16636273).
- **Amino-acid/nucleotide metabolism** — the interconverted folate pool feeds methionine (via methyl-THF and methionine synthase), histidine, and thymidylate synthesis (UniProt keywords: Methionine/Histidine biosynthesis; One-carbon metabolism).

Essentiality and phenotypes (ortholog evidence). In *E. coli*, FolD is essential; both of its activities are required to rescue a ΔfolD strain, and loss produces "formate and glycine auxotrophies" and altered antifolate/UV susceptibility (Aluri et al., 2016, P 26531681).
FolD is widely conserved and "involved in the biosynthesis of folate cofactors that are essential for growth and cellular development" (Eadsforth et al., 2012,

P 23050773).
In some bacteria (e.g., *C. perfringens*) the two activities are split between separate monofunctional proteins (FolD dehydrogenase + FchA cyclohydrolase) with Fhs providing an alternative 10-formyl-THF route (P 26531681) — but in *P. putida* FolD2 retains both activities in one polypeptide per HAMAP annotation.

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5. Structural Basis

FolD enzymes are two-domain proteins: an N-terminal folate/substrate-binding domain (Aminoacid_DH-like N; IPR046346) and a Rossmann-like **NAD(P)-binding domain** (IPR036291/IPR020631) that positions $NADP^+$. The active site lies in the interdomain cleft. High-resolution crystal structures of closely related Pseudomonad and Acinetobacter FolD confirm this architecture: "The crystal structure of *P. aeruginosa* FolD was determined at 2.2 Å resolution … for modelling of ligand complexes as well as for comparisons with the human enzyme" (Eadsforth et al., 2012, P 22558288).
The strong structural conservation between bacterial and human FolD/MTHFD1 has been noted as a challenge for selective antibacterial inhibitor design (P 22558288).
FolD2 (284 aa, 30.6 kDa, homodimer) conforms to this conserved fold.

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6. Supported and Refuted Hypotheses

Supported - FolD2 is a bifunctional $NADP^+$-dependent 5,10-methylene-THF dehydrogenase / 5,10-methenyl-THF cyclohydrolase (EC 1.5.1.5 + 3.5.4.9). *(UniProt catalytic annotation; ortholog biochemistry PMIDs 26531681, 25988590.)* - Its product 10-formyl-THF feeds purine synthesis and fMet-tRNA formylation. *(Keywords; P 16636273.)*

- It is a soluble cytoplasmic homodimer with a conserved two-domain fold. *(UniProt SUBUNIT; structures PMIDs 22558288, 23050773.)* - FolD2 is the canonical close FolD ortholog (86.6% identical to structurally solved *P. aeruginosa* FolD) with all seven key catalytic residues (Y50/K54/C58/Q98/D121/G122/R191) conserved — including cyclohydrolase-specific K54 and dehydrogenase-specific R191 — confirming an intact, genuinely bifunctional active site. *(Own sequence analysis; PMIDs 22558288, 25988590.)*

Refuted / cautioned - FolD2 is NOT NAD^+-specific and NOT membrane-associated. - Essentiality of *folD2* specifically in *P. putida* should NOT be assumed: because a paralog (*folD1*) exists, folD2 loss may be buffered — an inference from gene copy number, not a knockout in this strain.

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7. Limitations and Future Directions

- No *P. putida*-specific biochemical or knockout study of FolD2 was found; functional assignment rests on strong homology (HAMAP MF_01576) and well-characterized orthologs.
- The division of labor between *folD1* and *folD2* (redundancy, differential regulation, or kinetic specialization) is unresolved. The two paralogs share only 53.5% identity; FolD2 is the closer match to canonical FolD (86.6% to *P. aeruginosa*), suggesting FolD2 is the principal housekeeping copy while FolD1 may be specialized. Suggested experiments: single/double knockouts with growth/auxotrophy tests, RNA-seq under one-carbon-limiting conditions, and enzyme kinetics of purified FolD1 vs FolD2.
- No *P. putida* transposon-essentiality or CRISPRi dataset specific to PP_2265 was located in the searched literature; its individual essentiality (given FolD1) remains an open experimental question.
- Direct localization (fractionation) and structure determination for FolD2 would confirm the inferred cytoplasmic, homodimeric, NADP-dependent properties.

Key References

- Aluri et al. 2016, *FEBS J*-type study, PMID **26531681** — bifunctional FolD physiology; ΔfolD rescue and auxotrophies.
- Sah & Varshney 2015, PMID **25988590** — catalytic residues separating dehydrogenase vs cyclohydrolase activities.

- Eadsforth et al. 2012 (*P. aeruginosa* FolD), PMID **22558288** — 2.2 Å crystal structure; drug target.
- Eadsforth et al. 2012 (*A. baumannii* FolD), PMID **23050773** — conservation, essentiality, inhibitor complexes.
- Nilsson et al. 2006, PMID **16636273** — folD/fmt link to initiator-tRNA formylation.
